

Vitamins & Minerals — Selection Criteria

Screened roughly 930 multivitamins. All failed the criteria below.

Decided to buy vitamins and minerals separately instead.

Strict criteria — reasons to reject a product

Constraint	Description
Vitamin A (as <i>retinyl palmitate</i> , <i>retinyl acetate</i>)	Ready-made vitamin A cannot be excreted and builds up in the liver. Beta-carotene is converted only as needed — but isolated high-dose beta-carotene raised lung cancer risk in smokers and ex-smokers (ATBC and CARET trials). Safe only as food, or as a mixed carotenoid complex within limits. No multivitamin does this correctly. Limit 3,000 µg/day. NIH
Vitamin B6 (as <i>pyridoxine HCl</i>)	High long-term doses are linked to nerve damage in hands and feet. Reported cases involve mostly this form, not the active form (<i>P-5-P</i>). EU limit 12 mg/day, US 100 mg/day. EFSA 2023
Folate (as <i>folic acid</i>)	Folic acid is synthetic and must be converted by a liver enzyme that works slowly in humans. Unconverted folic acid then circulates in the blood. The active form (<i>L-5-MTHF</i>) skips this step. Limit 1,000 µg/day. PNAS study · NIH
Iron (as <i>ferrous fumarate</i> , <i>ferrous sulfate</i>)	Humans have no way to excrete iron, so it accumulates. Men and non-menstruating adults rarely need extra. Wanted zero or trace only. NIH
Zinc (as <i>zinc oxide</i>)	Cheapest form and the least absorbed. Total intake above 25 mg/day lowers copper status. NIH
Copper (missing, or as <i>copper oxide</i>)	Zinc displaces copper in the gut, so zinc without enough copper leads to copper deficiency. Many products omit copper entirely, or use an oxide that is barely absorbed. Want at least 0.9 mg in an absorbable form. NIH
Titanium dioxide (<i>E171</i> , tablet coating)	Banned as a food additive in the EU since 2022 after safety concerns over genetic damage. Found only in coated tablets, never in capsules. EFSA 2021
Component in several forms (complexes, proprietary blend)	Vendors are not obliged to state the weight or percentage of each form, and the listed order means nothing. “Magnesium 100 mg (bisglycinate, oxide)” can be 99% oxide and 1% bisglycinate. Without per-form figures the product cannot be checked.

Not a concern — covered by diet, or bought separately

Nutrient	Source	Nutrient	Source
Vitamin D	Skin makes none Oct–Mar in northern Europe. Buy separately — cheap, sold everywhere.	Potassium	Food only. Supplements capped near 99 mg — 3% of daily need. Potatoes, beans, greens.
Vitamin C	Fruit and vegetables. Cheap to buy separately.	Choline	One or two eggs give a third to half the daily 550 mg.
Vitamin E	Nuts, seeds, vegetable oils. Deficiency very rare.	Calcium	Dairy, leafy greens, tinned fish with bones.
Vitamin K	K1 from leafy greens; K2 from fermented food and gut bacteria. Deficiency very rare in adults.	Phosphorus	Meat, dairy, grains, legumes. Deficiency essentially unknown.
Beta-carotene	One carrot supplies 5–8 mg.	Iodine	Iodised salt, fish, dairy.
Lutein, zeaxanthin	Egg yolk (best absorbed), kale, spinach.	Sodium, chloride	Salt.

What was bought instead

Product	Dose	Key forms
Thorne, Basic B Complex 60 capsules	1/day	All eight B vitamins. B6 10 mg (as <i>P-5-P</i>), folate 400 µg (as <i>L-5-MTHF</i>), B12 400 µg (as <i>methylcobalamin</i>), B2 (as <i>riboflavin-5-phosphate</i>). No stearates.
NOW Foods, Full Spectrum Minerals 240 capsules	2/day	Copper 1 mg (as <i>bisglycinate</i>), zinc 7.5 mg (as <i>bisglycinate</i>), selenium 50 µg (as <i>selenomethionine</i>), manganese 2.5 mg, magnesium 250 mg. No iron.

Daily upper limits (stricter of EU / US)

Vitamin	Limit	Body	Mineral	Limit	Body
Vitamin A	3,000 µg	EFSA / US	Zinc	25 mg	EFSA
Vitamin B6	12 mg	EFSA	Copper	5 mg	EFSA
Folic acid	1,000 µg	EFSA / US	Selenium	255 µg	EFSA
Beta-carotene	3.5 mg	BfR (DE)	Manganese	11 mg	US

Upper limits cover total daily intake including food, not supplements alone. Beta-carotene figure is a German BfR recommendation; no binding EU limit exists. Not medical advice.